

DA2402:Data Curation and Visualization

Dr. Arun B Ayyar

Dr. Nirav Bhatt

Data Science and AI Dept, IITM

Course logistics

- Instructor: Dr. Arun B Ayyar
 - Office hours: Will be updated
- Teaching assistants: Office is still deciding
 - Office hours: Will be updated
- Course page and announcements: Moodle
- Course materials : Github : <https://github.com/oldmonkABA/da2402>
- Classes: Wednesday and Thursday. Makeup slot: Monday.

Assessment Method

- Assignments — 10
 - 2 assignments, roughly one around the **end of Aug**, second at the **end of Sept**
- Mid Sem — 25 (open notes)
 - **16-September-2026**, will contain whatever we cover till then
- Project — 25
 - Choice of project : Start of September '26 (list will be shared/ you can choose your own)
 - Presentations : **Week of 26 – October -2026**
- End Sem — 40 (open notes)
 - **11-November-2026**, covers everything

The project — 25% of your grade

- Teams of 4. Start thinking about datasets around the end of August.
- The brief: take a real, messy dataset and carry it end to end.
 - Acquire it — scrape, API, or a public source with a real license
 - Curate it — clean, document, justify every decision you made
 - Visualize it — charts that answer a question you actually posed
 - Ship it — an interactive dashboard someone else can use
- Deliverables: cleaned dataset + notebook + dashboard + presentation
- Graded on: curation rigour, chart choice, dashboard usability, communication

Policy on AI tools

- Allowed:
 - Explaining concepts, syntax, error messages
 - Debugging code you wrote yourself
 - Rules for Project:
 - You can use LLMs for coding up project.
 - **However, You must be able to explain any question that is asked on the code**
- Not allowed:
 - Submitting generated code or analysis as your own work for Assignments.
 - **Use of LLMs during mid-sem and end-sem is strictly prohibited and will invite disciplinary action (Last year 2 students got U)**

The course outline

- **Part I — Curation**

- 1. Introduction to data curation
 - Why it matters, acquisition, lifecycle, governance
- 2. Data cleaning methods
 - Data types, quality issues, missing data, outliers
- 3. Data wrangling techniques
 - Groupby, pivot, reshaping, integration, time series

- **Part II — Visualization & dashboards**

- 4. Introduction to data visualization
 - Principles, chart types, libraries, grammar of graphics
- 5. Standard visualization charts
 - Distributions, frequencies, trends
- 6. Advanced charts
 - Hierarchies, maps, text, time series
- 7. Interactive dashboards in Plotly-Dash
- 8. Gestalt Principles

What this course is not

- Not a machine learning course
 - No model training, no algorithms, no backprop
- Not a statistics theory course
 - We use means, correlations and distributions. We don't derive them.
- Not a graphic design course
 - Though taste matters, and we will argue about it
- What it is:
 - Getting messy real-world data into a state you can trust
 - Then making it say something to someone who isn't you

Background needed

- Coding
- Python
- Pandas
- Statistics
- Database fundamentals

Tools — install these before next class

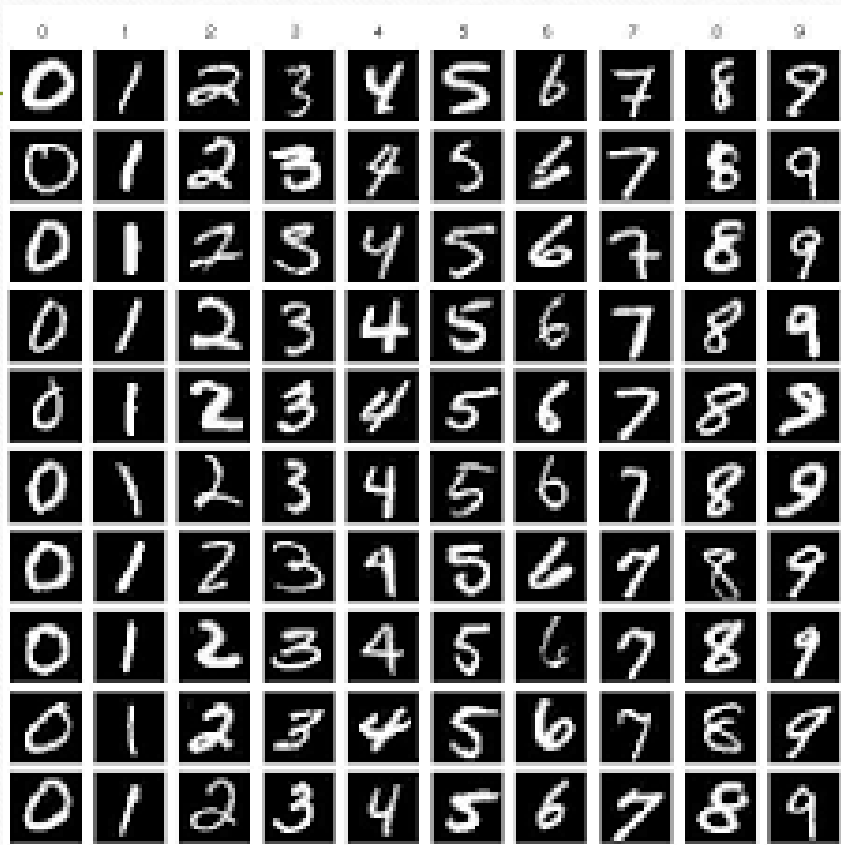
- **Software**

- Python 3.11 or later
- Jupyter Notebook / JupyterLab
- VS Code
- Plotly-dash
- Git, and a GitHub account

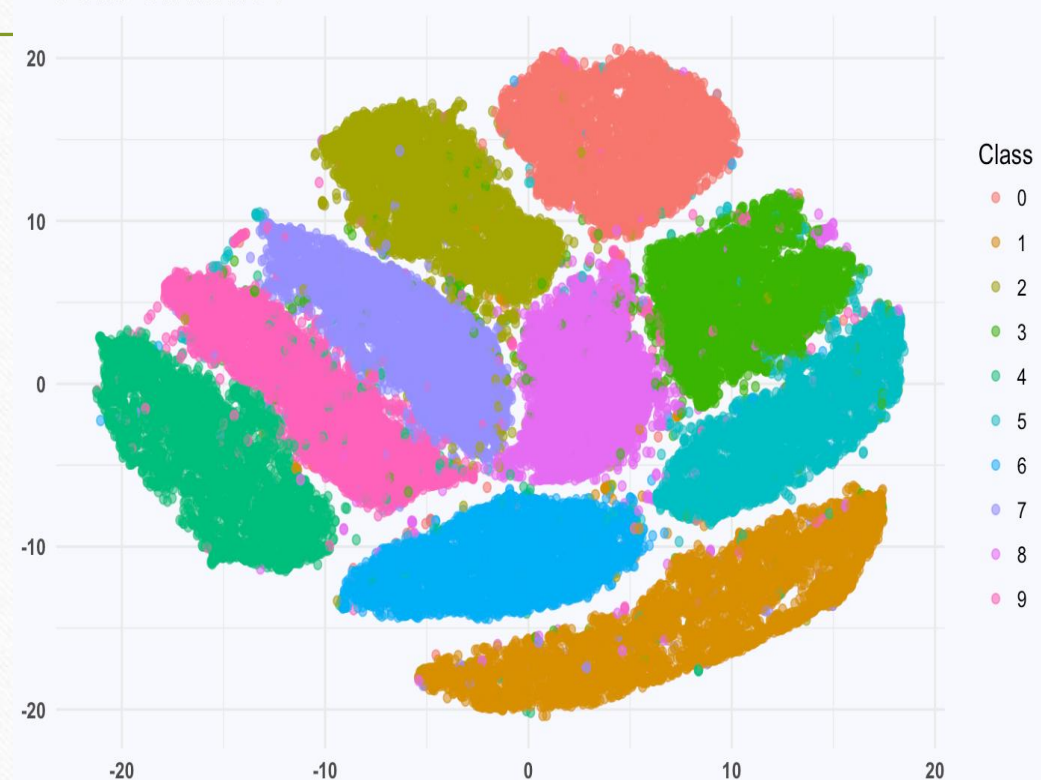
- **Python libraries**

- pandas, numpy
- matplotlib, seaborn
- Plotly-dash
- `pip install pandas numpy matplotlib seaborn plotly dash`
- Bring a working laptop to every class.

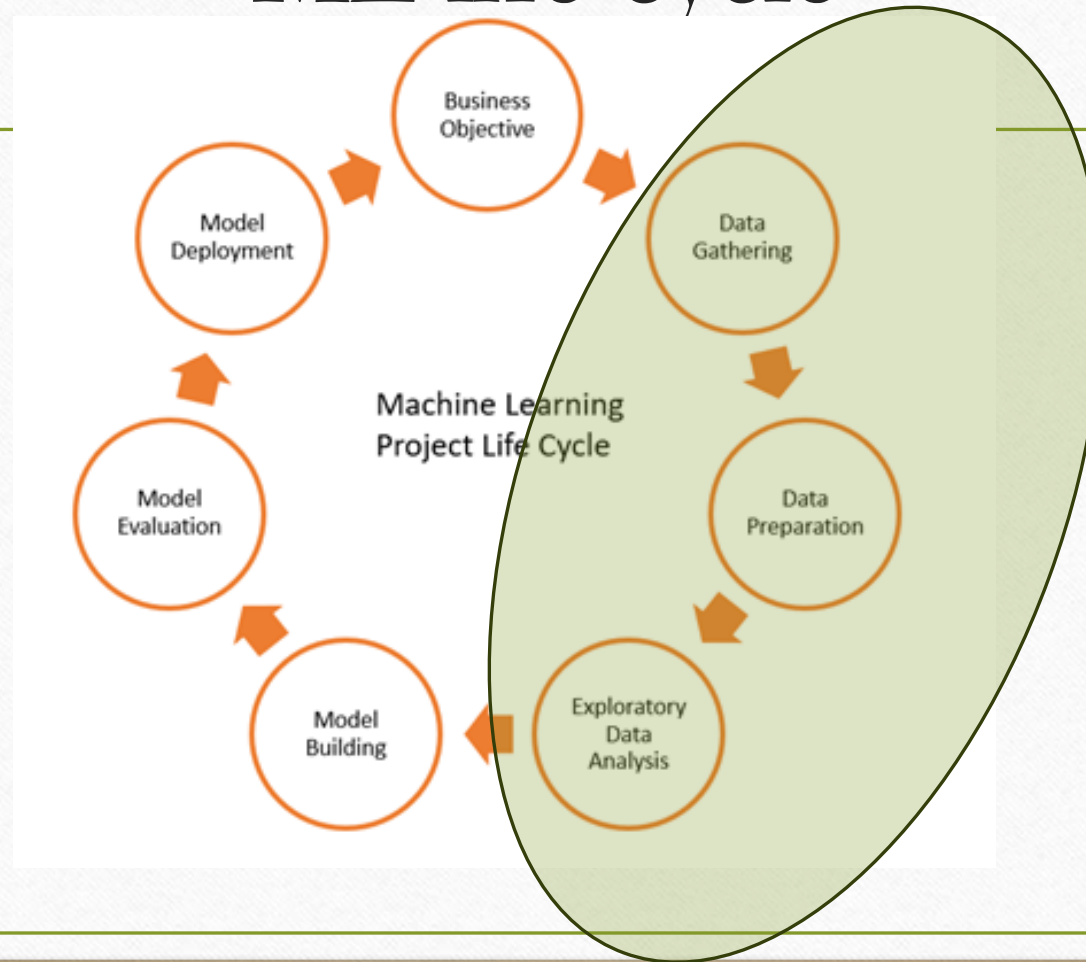
MNIST data in lower dimension



t-sne on MNIST



ML life cycle



The data lifecycle

- Create / acquire — where did it come from, and under what terms?
- Store — format, schema, versioning, backups
- Process — cleaning and transformation, recorded rather than remembered
- Analyze & visualize — what the rest of this course builds toward
- Share / publish — for whom, and at what granularity?
- Archive / dispose — retention limits are often legal, not optional

What "curation" actually means

- Cleaning is one link in a longer chain:
 - Acquire → Describe → Clean → Wrangle → Integrate → Store → Govern → Preserve
- Wrangling (Modules 2 and 3) is the part you'll spend most keystrokes on.
- It is not the part that makes data trustworthy.
- The test for whether you curated something:
 - Can someone else pick up this dataset in two years and rely on it?
 - That needs the data, plus metadata, provenance, and every decision you made
- "I dropped the nulls" is not a decision anyone can audit.
- "I dropped 412 rows with missing timestamps, listed here, because ..." is.

What bad curation costs

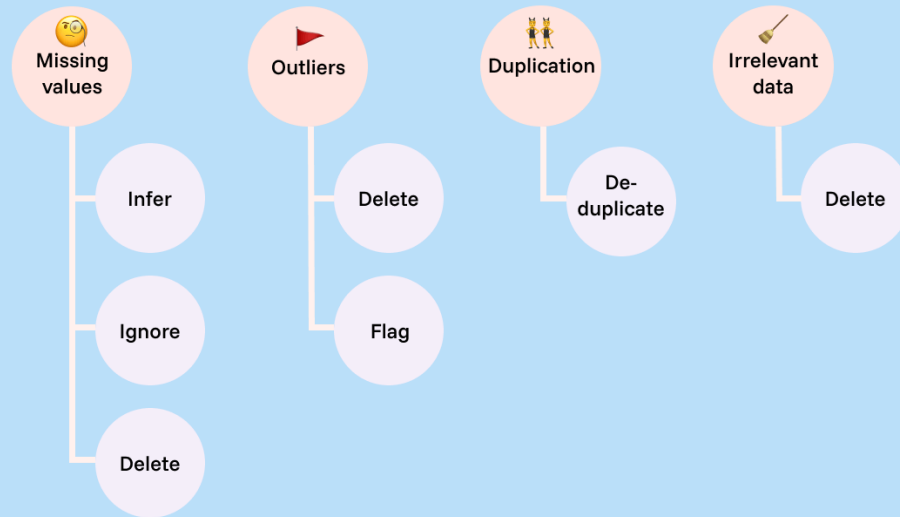
- Public Health England, October 2020
 - Case data was collated in an old .xls file, which caps at 65,536 rows
 - Rows past the limit were silently dropped — about 16,000 COVID cases
 - Thousands of contacts were never traced. A file format did that.
- Mars Climate Orbiter, 1999
 - One team worked in pound-force seconds, the other in newton-seconds
 - The spacecraft entered the atmosphere too low and was lost. \$125M.
- Survey after survey puts 60–80% of an analyst's time on finding and cleaning data.
- None of these were modelling failures. Every one was a curation failure.

Governance & compliance

- PII — anything that identifies a person, directly or in combination
 - Name and phone number, obviously
 - But also: pin code + date of birth + gender identifies most people
- India — Digital Personal Data Protection Act, 2023
 - Consent, purpose limitation, rights of the data principal
- EU — GDPR
 - Same themes, extraterritorial reach, penalties with real teeth
- Practical rules for this course:
 - Check the licence before you use a dataset — "it was on the internet" is not a licence
 - Read robots.txt and the terms of service before you scrape
 - Anonymize before you share, and assume re-identification is easier than you think

Data Wrangling

The data cleaning process



- Data Collection (web scraping / files / api)
- Data Cleaning
- Understanding relation between various variables

What dirty data actually looks like

customer_id	name	signup_date	city	age	spend_inr
1001	Ravi Kumar	2024-03-11	Chennai	34	12500
1002	ravi kumar ¹	11/03/2024 ²	chennai ³	34	12,500 ²
1003	Priya S	2024-13-45 ⁵	Bengaluru	29	8400
1004	— ⁴	2024-04-02	Mumbai	-7 ⁵	9100
1005	Arjun Mehta	2024-04-15	Bombay ³	41	— ⁴
1006	Sara Ali	2024-05-01	Delhi	38	9999999 ⁶

¹ **Duplicate record**

1002 is 1001 again, typed differently

² **Inconsistent format**

two date formats; a number stored as text

³ **Inconsistent categories**

chennai / Chennai; Mumbai / Bombay

⁴ **Missing values**

no name, no spend — and they mean different things

⁵ **Invalid / impossible**

month 13, day 45; a negative age

⁶ **Outlier**

₹99,99,999 — real high-roller, or a data-entry slip?

Types of data

- Tables
- Image
- Text
- Speech
- Time Series
- Geodata

Every kind of data gets dirty differently

- Tables — the six you just saw: missing, duplicate, inconsistent, invalid, outlying
- Text
 - Encoding mojibake, stray HTML, inconsistent casing, near-duplicate strings
 - "IIT Madras", "IIT-Madras", "IITM" — three spellings, one entity
- Time series
 - Gaps, irregular sampling, duplicated timestamps, timezone and DST confusion
 - A missing hour and an hour of zeroes mean opposite things
- Geodata
 - Swapped lat/long, wrong projection, place names that moved or were renamed
 - The infamous 0°N 0°E — Null Island, where failed geocoding goes
- Images & speech
 - Corrupt files, mixed resolutions, duplicated frames, mislabelled classes

Why cleaning is important

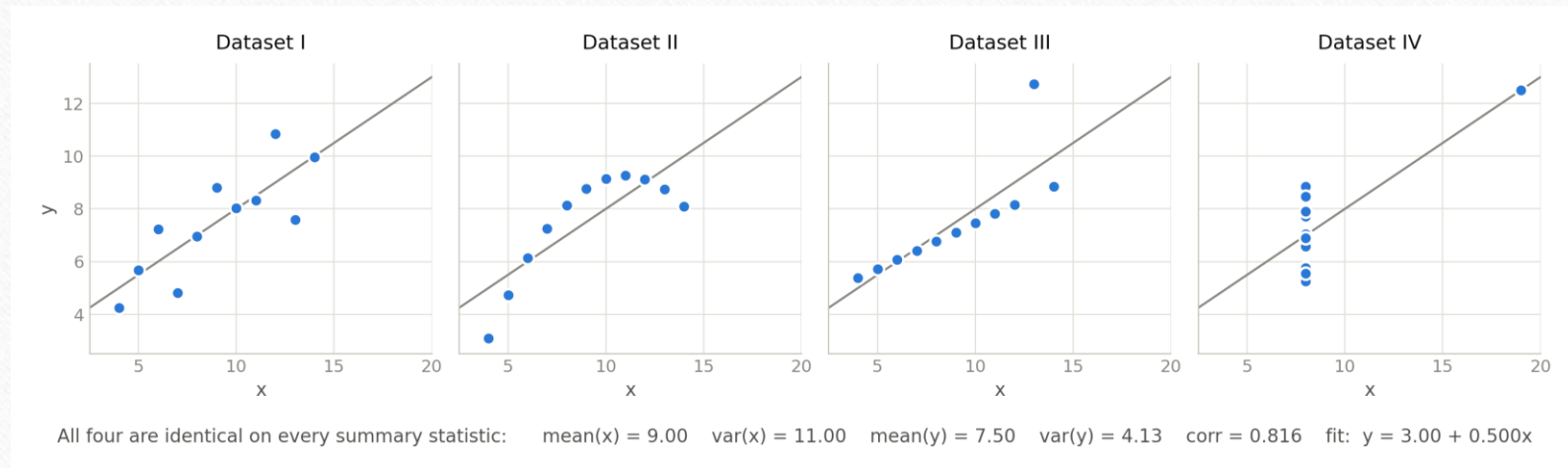
- Every cleaning decision is a claim about the world.
- A missing value can mean at least three different things:
 - Not collected · not applicable · collected and genuinely zero
 - Dropping all three the same way throws away the difference
- An outlier is not automatically an error.
 - It might be your most important customer, or a sensor that failed
 - Deleting the ones that spoil the trend is how analysis becomes fiction
- The same table cleaned by two people gives two different answers.
- So the rule for this course:
 - Never clean silently. Record what you changed, how many rows, and why.
 - A cleaning step you cannot justify is a cleaning step you should not make.

Data Visualization

- Picture is worth a 1000 words
-

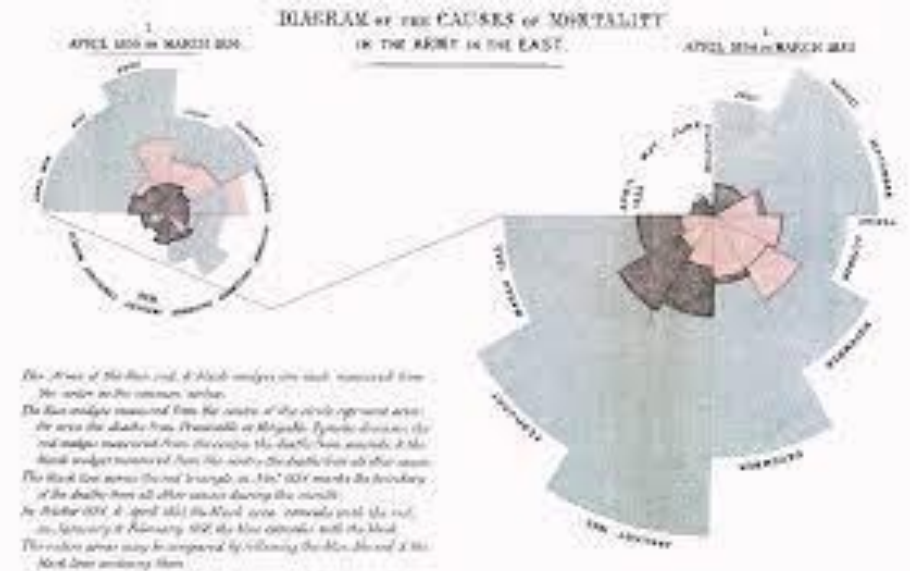
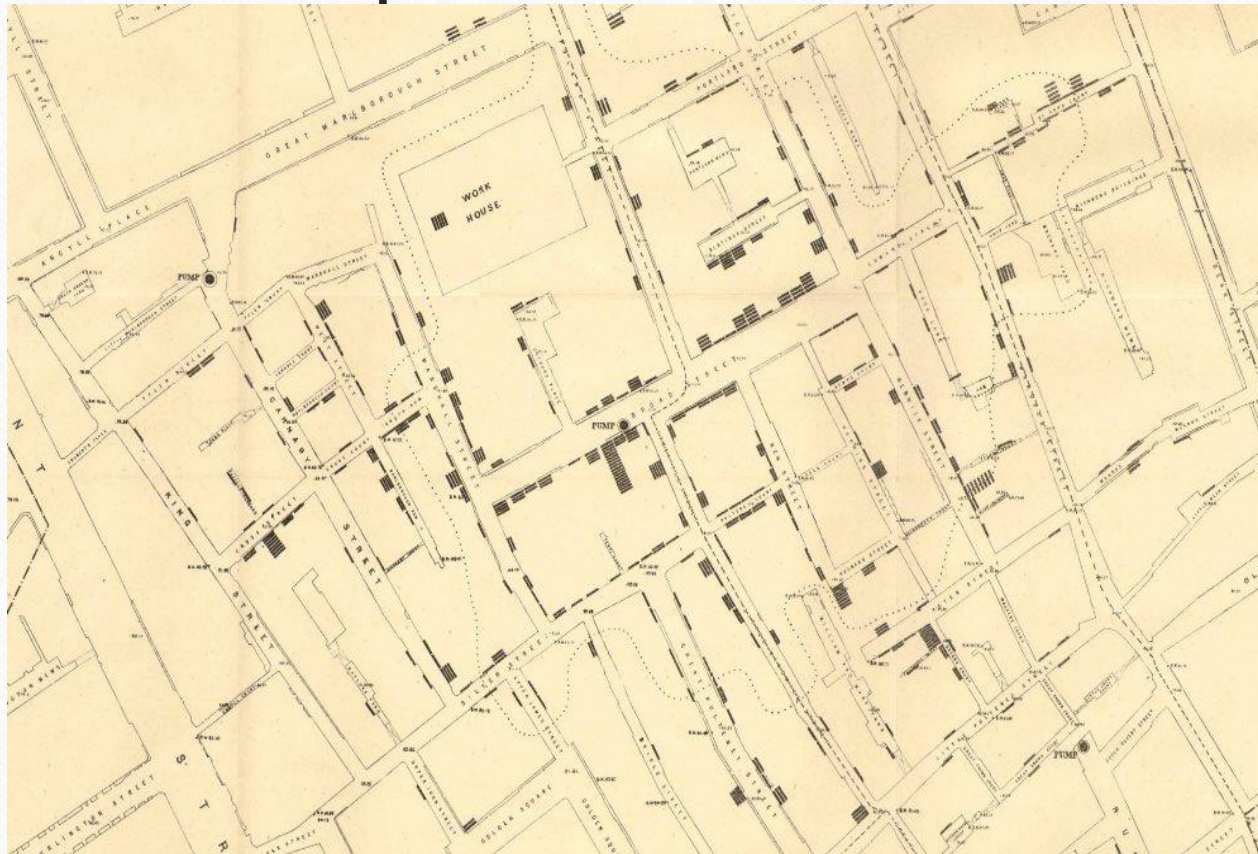


Why plot at all? Anscombe's quartet, 1973



Same mean, same variance, same correlation, same regression line — and four completely different datasets. Summary statistics cannot tell you what your data looks like. Plot it first, every time.

Impactful visualizations in the history

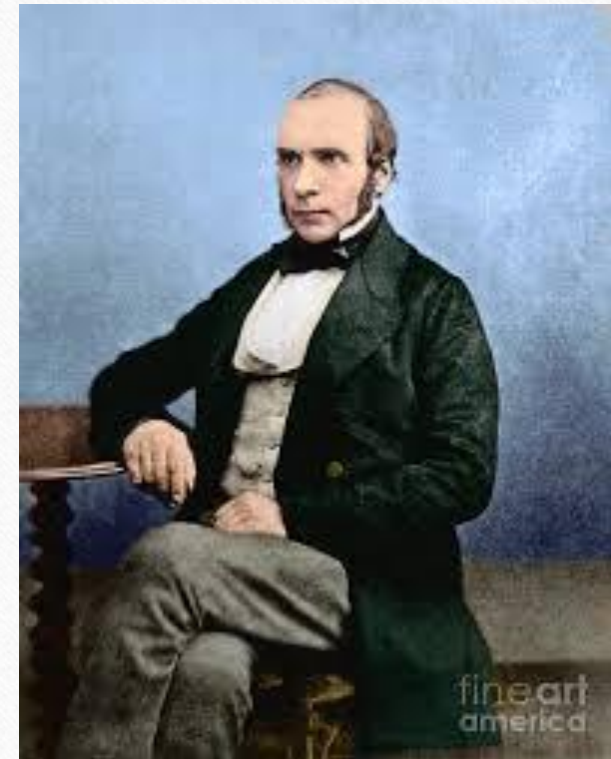


Viz 1

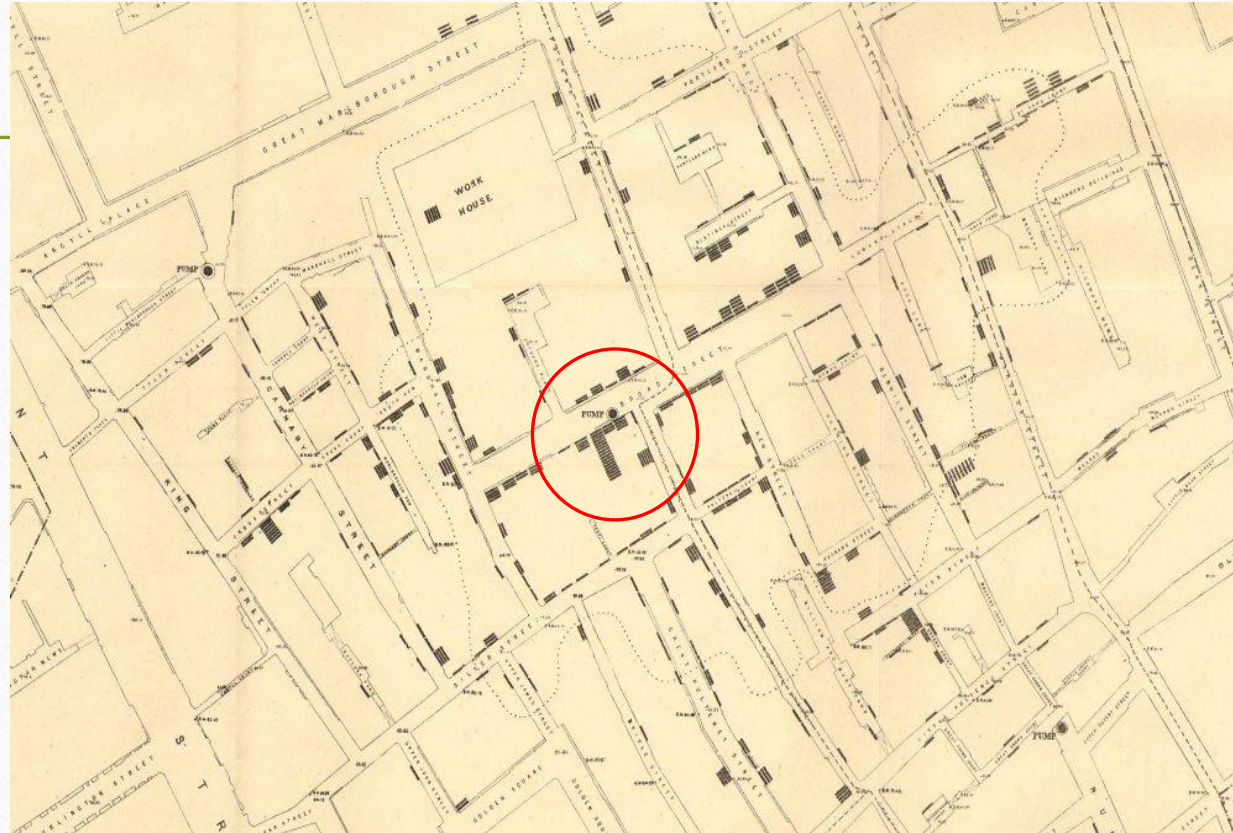
- The 1854 cholera outbreak in Soho, London, resulted in approximately 600 deaths in a 10-day period.
- The outbreak, centered around Broad Street (now Broadwick Street)
- Existing theories

 - In France, it was widely believed to be associated with poverty
 - In Russia, it was believed to be spread by contagion, but with an unknown mechanism
 - In the United States, the belief was that cholera had been brought by Irish immigrants.
 - In England, and London particularly, the most common view was that the disease arose from breathing the foul air around the River Thames, which stemmed from the nearly universal practice of discharging untreated sewage directly into the river. This theory was given a name: *miasma*, from the Greek word meaning “pollution,” came to mean poisonous air or vapors filled with particles from rotting organic matter.
- *Miasma* was a beautiful theory, but in ten years it would be destroyed by data and graphs.

Enter John Snow



John Snow's map — Broad Street, Soho, 1854



Each black bar is one cholera death, stacked at its address. The Broad Street pump sits at the centre of the cluster.

Analysis

- “The most terrible outbreak of cholera which ever occurred in this kingdom, is probably that which took place in Broad Street, Golden Square, and the adjoining streets, a few weeks ago.”
- “Within two hundred and fifty yards of the spot where Cambridge Street joins Broad Street, there were upwards of **five hundred** fatal attacks of cholera in ten days.”
- “On proceeding to the spot, I found that nearly all the deaths had taken place within a short distance of the pump. There were only ten deaths in houses situated decidedly nearer to another street pump.”
- Impact - . His findings resulted in a fundamental change to the design of the water and waste systems of London, and subsequently in other cities, which led to significant improvements in public health around the world.

Vis 2– Florence Nightingale

- **1854 – Britain enters the Crimean War**

- British soldiers are sent to fight in Crimea.
- Conditions in military hospitals are **unsanitary and overcrowded**.

- **Late 1854 – Nightingale arrives in Scutari**

- Florence Nightingale and her team of nurses begin working in the military hospital.
- They find that **disease, not wounds, is killing most soldiers**.

- **1855 – Deaths reach peak levels**

- In early 1855, **disease-related deaths spike dramatically**.
- Cholera, typhus, and dysentery are rampant.
- This is the period shown in the **first half of the rose diagram** with **large blue areas**.

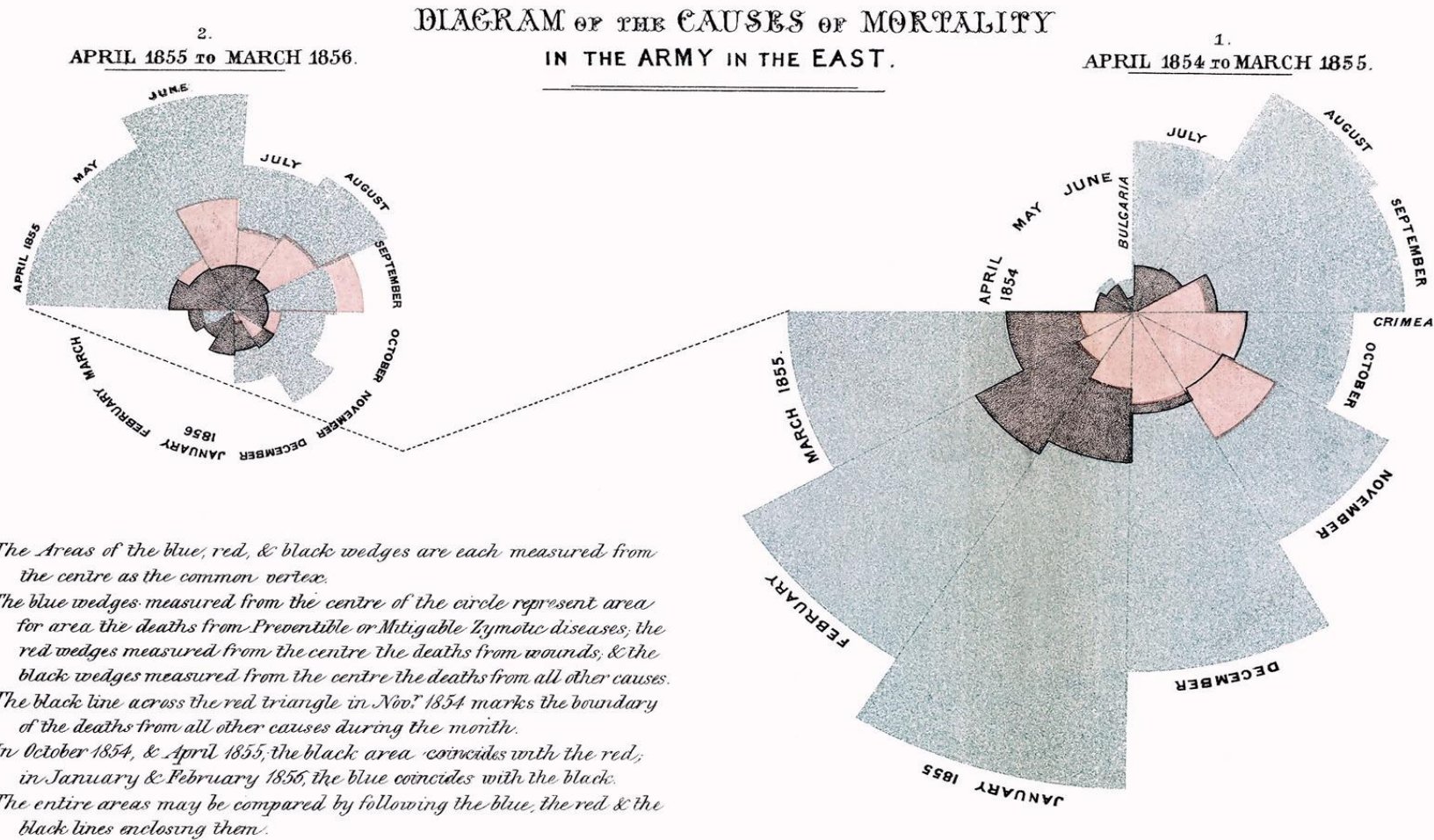
- **Spring 1855 – Sanitary Commission intervenes**

- British government sends a sanitary commission.
- Major reforms: **clean water, drainage, ventilation, and sewage systems** are introduced.

Women in Data Vis – Florence Nightingale

- **Mid-to-late 1855 – Mortality drops sharply**
 - Deaths from disease **decline drastically**.
 - This is reflected in the **second half of the rose diagram**, with **smaller blue areas**.
- **1856 – Crimean War ends**
 - Nightingale returns to England but continues analyzing military health records.
- **1858 – Rose Diagram is published**
 - Nightingale publishes her "**Diagram of the Causes of Mortality**".
 - It visually proves that **improved hygiene saved lives**, influencing public health and military policy.

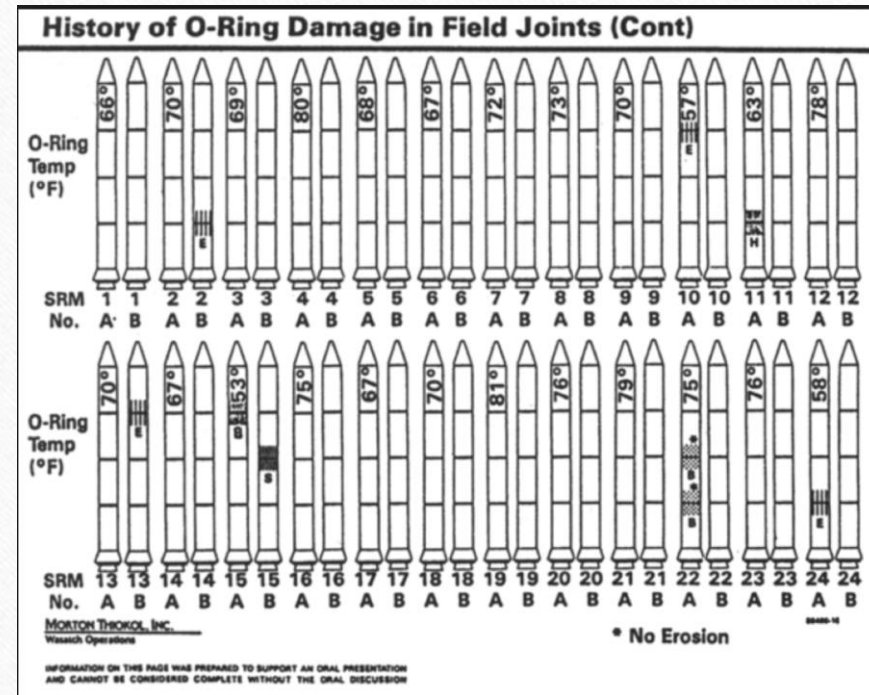
Nightingale's Rose Diagram, 1858



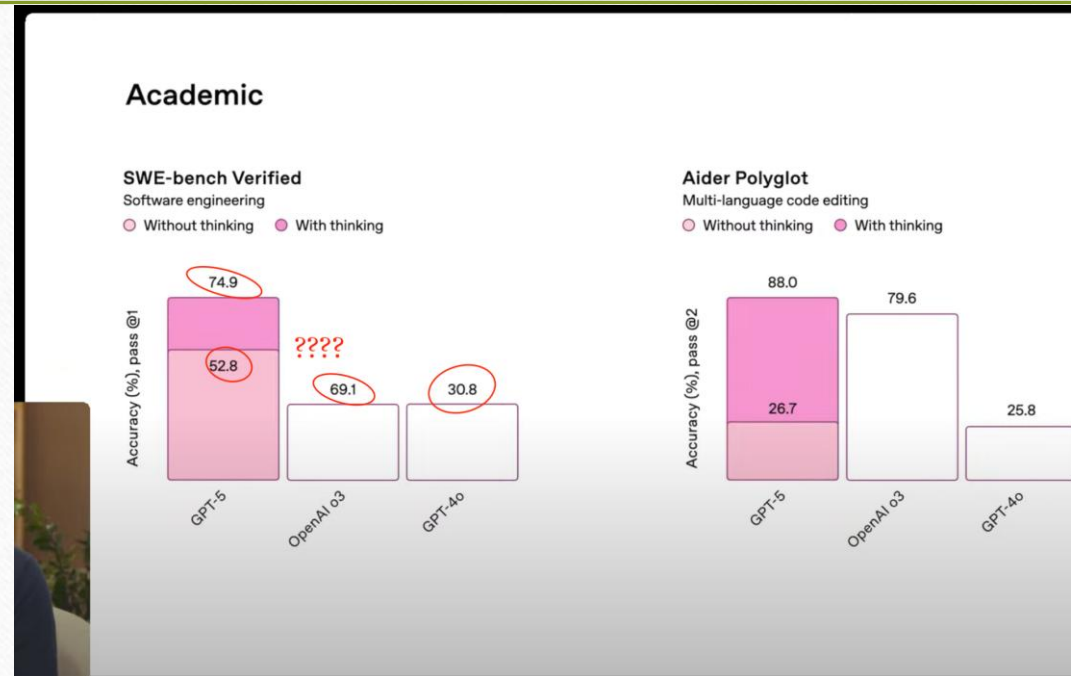
Blue = preventable disease. Red = wounds. Black = all other causes. Right panel: Apr 1854 – Mar 1855, before the Sanitary Commission.

When visualization fails — Challenger, 1986

- The night before launch, engineers argued the O-rings would fail in the cold.
- They had the data. They were right. They sent 13 charts.
- What the charts showed:
 - O-ring damage indexed by rocket motor serial number
 - Flights with no damage were left out of the plot entirely
 - Temperature appeared, but not as the organizing axis
- Plot damage against launch temperature, every flight included:
 - the relationship is impossible to miss
- The launch went ahead at 2°C. Seven people died.
- The data was there. The encoding was wrong.



How charts mislead



Truncated axes · dual axes · 3D and pie · cherry-picked date ranges · uneven bins · wrong aggregation.

Interesting Visualisations

- Hans Rosling / Gapminder
 - 200 countries, 200 years, four minutes — worth watching
- Our World in Data
 - The direct descendant of what Snow and Nightingale were doing
- COVID dashboards, 2020
 - Johns Hopkins' dashboard became public infrastructure in about a week
 - It was, underneath, a data curation problem long before a design one
- The graphics desks at NYT, FT, Reuters, Bloomberg
 - Data journalism is a career now, and it hires people who can do both halves
- The tools changed. The job did not: find what is true, then make it visible.

What you'll be able to build by December

- Scrape data from any internet source
- Curate the data from raw source to documented and trustworthy
- Interactive charts in plotly
- Principles to make the dashboard look neat.

Three questions we'll keep asking

- Univariate — what does one variable look like on its own?
 - Distribution, centre, spread, skew, outliers
 - Histograms, box plots, density plots
- Bivariate — how do two variables move together?
 - Relationship, strength, direction, and whether it is real
 - Scatter plots, grouped bars, line charts
- Multivariate — what happens when three or more interact?
 - Structure, clusters, hierarchies, confounds
 - Small multiples, treemaps, heatmaps, maps
- Every chart in Modules 5 and 6 answers one of these three.

Before next class

- Install everything on the tools slide, and check it imports

Search for the following:

- Look up John Snow's original map at full resolution — it rewards zooming
- Watch Hans Rosling, "200 Countries, 200 Years, 4 Minutes"